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Study on the microscopic damage evolution and dynamic fracture properties of sandstone under freeze-thaw cycles

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ABSTRACT

Freeze-thaw has great deterioration effect on rock mechanical performance, which strongly affect the stability of rock masses engineering. Meanwhile, natural fractures in real rocks are ubiquitous with considerable variations in size, number, and orientation, which further accelerate the risk of geological disasters. In this study, the microstructure changes of specimens induced by cyclic freeze-thaw were measured using nuclear magnetic resonance (NMR) technique. Dynamic fracture tests were performed using single cleavage triangle (SCT) red sandstone specimens. Crack propagation gauges (CPGs) were applied to determine the crack velocity. Meanwhile, the microstructure of the fracture surface was obtained with the aid of scanning electron microscope (SEM). The results show that the cyclic freeze-thaw plays an important part in rock dynamic fracture behavior. The porosity and crack velocity increase with freeze-thaw cycles, whereas the crack initiation time and dynamic fracture toughness decrease with freeze-thaw cycles. Sandstone has three pore types and is susceptible to the freeze-thaw weathering. The fracture surface of specimen without freeze-thaw is smooth and the fracture mode is transgranular. With the rise of freeze-thaw cycles, the fracture surface becomes rougher, indicating that intergranular fracture plays a dominant role in sandstone failure.

1. Introduction

Rock freezing and thawing weathering is a common and serious phenomenon in natural environment, which is induced by the deterioration of the internal structure of the rock masses under the change of environmental temperature. Freeze-thaw damage can strongly degrade rocks, which may induce serious geological disasters, e.g., rock falls, landslides and solifluction, etc. (Chen et al., 2019; Shen et al., 2021; Shen et al., 2018). Therefore, studying the changes in the internal structure and mechanical behavior of rocks under freeze-thaw conditions is very important, which can provide a great help to assess the engineering safety and to prevent geological disasters.

At present, non-destructive methods of ultrasonic detection (Khanlari et al., 2014), acoustic emission (Zhang et al., 2020), electrical resistivity (Chen and Lin, 2004), SEM (Wight, 2000), computed tomography (CT) (Suzuki et al., 2017; Zhang et al., 2004), and NMR (Yan et al., 2017) have been applied to evaluate the internal structure damage of rock materials. For example, Wang et al. (2016b) observed the changes in rock pore structure using SEM examination, and they found that the pores between mineral grains become larger and the cement decreases after freeze-thaw. Liu et al. (2019) revealed the damage process of rock induced by freeze-thaw using SEM examination and CT technique. Park et al. (2014) studied the expansion of micro-scale pore of three kinds of rocks produced from freeze-thaw experiments using CT technique, and they found that lithology has a considerable effect on freeze-thaw damage. Wang et al. (2020) investigated the freeze-thaw deterioration mechanism of rock containing two pre-cracks using acoustic emission technique and CT technique, and pointed out that the deterioration of rock is an accumulative freeze-thaw damage. Liu et al. (2016) used CT images to analyze the parameters such as lengths, widths, and areas of the pores inside the rock, and they found that the freezing process of water-filled pores is mainly related to pore

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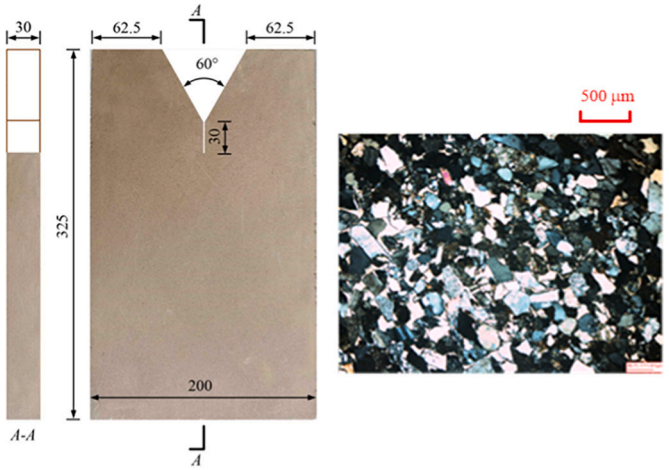


Fig. 1. Geometry of SCT specimen (unit: mm) and optical microscopic image.

size. Jia et al. (2019) defined the representative element of frost damage to assess the rock freeze-thaw damage with the aid of SEM examination and NMR technique. Compared with other non-destructive methods, NMR technique has the advantage of quantifying mesoscopic damage, and there is no radiation damage to the human body. Recently, NMR technique has been widely applied in the study of microscopic damage mechanism of rocks (Li et al., 2018; Li et al., 2016; Liu et al., 2018; Zhou et al., 2015).

In real rocks, natural fractures are ubiquitous with considerable variations in size, amount, and orientation (Li et al., 2019a; Wang et al., 2017; Yang et al., 2018). For example, Sagong and Bobet (2002) systematically studied the effects of nine types of cracks on rock mechanical properties. Feng et al. (2009) investigated the mechanical characteristics of rock under the effects of various factors including crack types, hydration environments and stress. Yang (2011) investigated the failure modes of rock containing two coplanar cracks, and they analyzed the relation between crack angle and fracture behavior. For fractured rock masses in cold environment, Huang et al. (2019) pointed out that the frost heaving force can cause the crack to initiate, and the angle and length of the crack have a significant effect on fracture characteristics. Han et al. (2016) and Kang et al. (2013) discussed the degradation effects of chemical solutions and cyclic freeze-thaw on rock. Abdolghanizadeh et al. (2020) reported the cyclic freeze-thaw can remarkably diminish the rock fracture toughness. Mu et al. (2017) investigated the freeze-thaw deterioration characteristics of three types of fractured rock, and they obtained the shear strength of three lithological rocks.

Although many scholars have performed the studies about the freeze-thaw damage characteristics of rocks, less attention has been paid to the dynamic fracture characteristics of rock under cyclic freeze-thaw. In this research, SCT red sandstone specimens were performed freeze-thaw experiment, the changes in pore structure of specimens were detected using NMR technique. Meanwhile, dynamic fracture experiments were conducted on the SCT specimens (Dong et al., 2018), and the dynamic failure process of specimen was monitored by CPG. Furthermore, the fracture surfaces of sandstone specimens were obtained by SEM technique, and the fracture modes of the sandstones were also analyzed.

2. Test methodology

2.1. Specimen and material

The rock material applied in experiment were uniform fine-grained red sandstones, the mineralogical contents of the red sandstone were detected by X-ray diffraction and OLYMPUS BX51 microscope, its mineral composition mainly includes 29% quartz (SiO_2), 13% analcime ($NaAlSi_2O_6{\cdot}H_2O$), 24% albite ($NaAlSi_3O_8$), 32% muscovite ($KAl_3Si_3O_{10}(OH)_2$), and 2% clay mineral. SCT specimen was selected to conduct the dynamic fracture test for its simple manufacture. Compared with other specimen configurations, the SCT test procedure is simple and does not require molds, such as three-point bending supports. The SCT dimension is sketched in Fig. 1, and the crack width is 1 mm.

To reduce the specimen heterogeneity effect on data quality, all specimens were derived from the same rock block, and specimens without visible flaws were selected first, then ultrasonic measurements were performed to select specimens with ultrasonic waves deviation within 3%. In addition, the top and bottom ends of the specimens were polished by an angle grinder, and the unevenness was within 0.05 mm.

2.2. Experimental procedure

Six groups were treated with 0, 5, 10, 20, 40, and 60 freeze-thaw cycles, respectively. All specimens were first dried at 105 °C to let the specimens be completely dry, then submerged into water for 7 days to

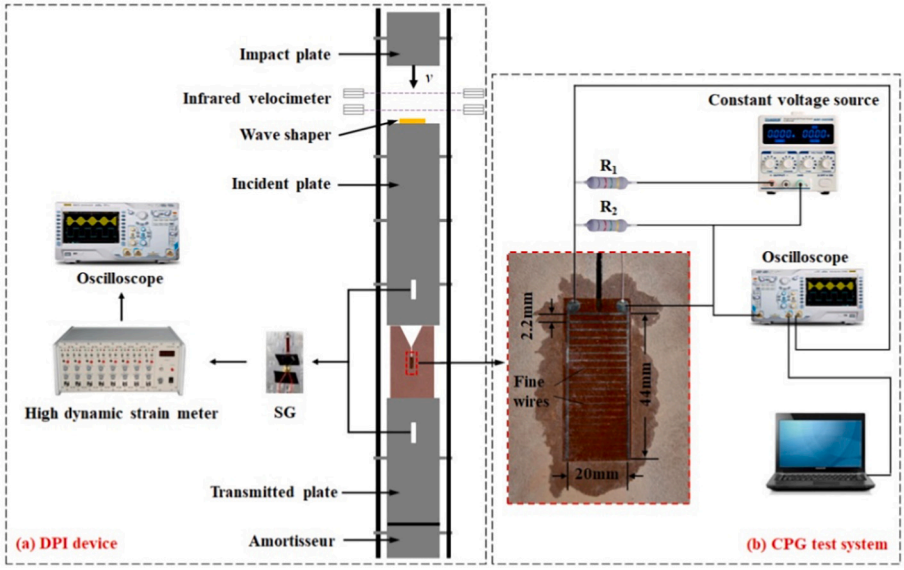


Fig. 2. DPI device and CPG test system.

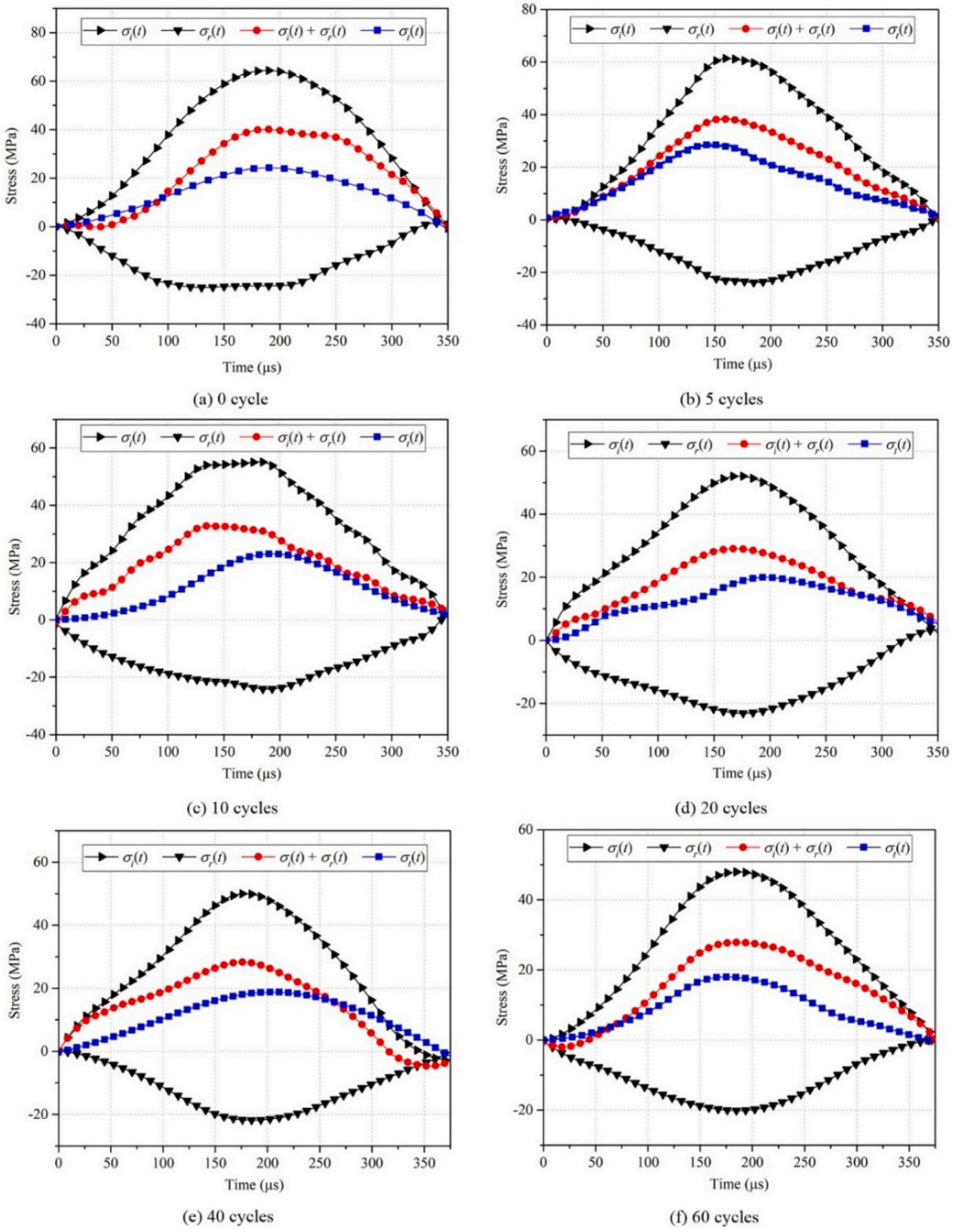


Fig. 3. Time-dependent loading curves for sandstone subjected to different freeze-thaw cycles.

get fully-saturated specimens (Niu et al., 2020). The water content of fully-saturated sandstone is 7.31%. After that, all specimens were placed in a freezing chamber at −20 °C for 12 h, and then melted in water at 20 °C for 12 h, thus, one cycle was completed. Finally, 4 specimens for each group, and totally 24 specimens were prepared. Ultrasonic measurements, NMR tests, SEM examinations, and impact tests were implemented for each group of sandstone specimens.

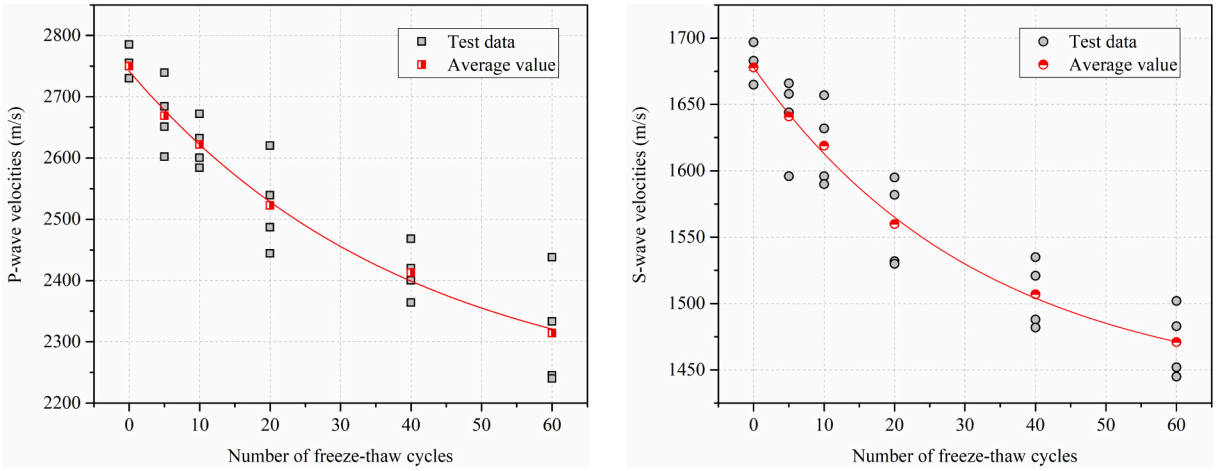
2.3. NMR technique

NMR can provide important information about rocks including the porosity, permeability, viscosity and pore size distribution. A 100%

Table 1
Parameters of sandstone at various freeze-thaw cycles.

Specimen ID	c_p /(m/s)	Ave. /(m/s)	c_s /(m/s)	Ave. /(m/s)	c_R /(m/s)	ρ /(kg/m^3)	E_d /GPa	μ_d
0–1	2785	2750	1697	1678	1599	2349	15.91	0.204
0–2	2755		1683					
0–3	2732		1670					
0–4	2728		1660					
5–1	2684	2669	1658	1641	1559	2355	15.17	0.196
5–2	2651		1611					
5–3	2602		1596					
5–4	2739		1666					
10–1	2632	2622	1632	1619	1536	2360	14.74	0.192
10–2	2600		1596					
10–3	2584		1590					
10–4	2672		1657					
20–1	2539	2523	1582	1560	1479	2357	13.65	0.191
20–2	2487		1532					
20–3	2444		1530					
20–4	2620		1595					
40–1	2420	2413	1521	1507	1423	2356	12.63	0.181
40–2	2400		1488					
40–3	2364		1482					
40–4	2468		1535					
60–1	2333	2314	1483	1471	1377	2343	11.77	0.161
60–2	2245		1452					
60–3	2240		1445					
60–4	2438		1502					

Note: 0–1 means the first specimen for group 0 freeze-thaw and so on.



(a) P-wave velocities (b) S-wave velocities

Fig. 4. Relations among P-wave velocities, S-wave velocities and freeze-thaw cycles.

water-saturated specimen is placed in a magnetic field, as the frequency of the hydrogen nuclei orbiting the external magnetic field is the same as that of the electromagnetic field, resonance occurs. When the applied magnetic field is removed, the hydrogen nuclei will release the absorbed energy signal, which can be measured and the T_2 transverse relaxation time distribution can be obtained (Coates et al., 1999).

There are three independent relaxation mechanisms in rock pores, i.e., bulk relaxation T_{2f}, surface relaxation T_{2s} and diffusion relaxation T_{2k}. The T_2 relaxation time can be expressed as:

$$\frac{1}{T_2}=\frac{1}{T_{2f}}+\frac{1}{T_{2s}}+\frac{1}{T_{2k}} \tag{1}$$

For water-saturated rocks, T_{2k} and T_{2f} are often negligible, and T_{2s} is a function of the ratio of pore surface (S) to volume (V):

$$\frac{1}{T_2}\approx\frac{1}{T_{2s}}=\rho_2\left(\frac{S}{V}\right)_{pore} \tag{2}$$

where ρ_2 denotes surface relaxivity, and S/V is related to pore size (r):

$$T_2=\frac{1}{\rho_2 F_s}r \tag{3}$$

where F_s is the pore shape factor. For spherical pores, $F_s = 3$, and for columnar pores, $F_s = 2$. Let $C = 1/\rho_2 F_s$, and the Eq. (3) can be further written as:

$$T_2 = C{\cdot}r \tag{4}$$

One can find from Eq. (4) that the pore distribution inside the rock can be obtained from the T_2 distribution. The smaller the pore size (r) is, the shorter the relaxation time (T_2) is.

2.4. Impact tests

The impact test was conducted on drop plate impact (DPI) device designed based on the principle of SHPB device. The DPI device is mainly composed of an incident plate, a transmitted plate, an impact plate, strain gauges, a high dynamic strain meter and an oscilloscope, as

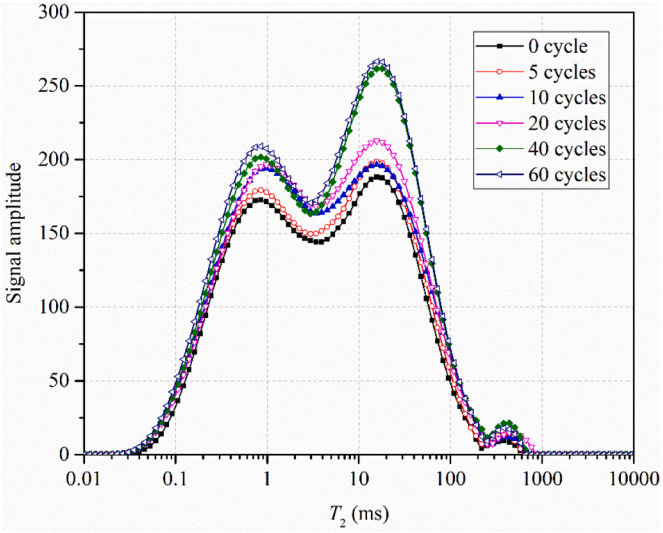


Fig. 5. T_2 spectrum curves of specimens at various freeze-thaw cycles.

illustrated in Fig. 2(a). Addtionally, a brass plate was applied to the experiment to eliminate wave oscillation.

To measure crack propagation parameters, i.e., crack initiation and propagation time, the CPG test system given in Fig. 2(b) was applied in the experiment. The system consists of a CPG, a constant voltage source and an oscilloscope. The CPG is composed of 21 fine wires with different widths but 20 mm in length identically, and the distance between two adjacent wires is 2.2 mm. In the study, impact tests using SCT specimens were conducted at the impact velocity 7.0 m/s.

3. Results and analysis

3.1. Time-dependent loading curves

According to Zhou et al. (2012), the stress applied on specimen top end (σ_{top} (t)) and bottom end (σ_{bot} (t)) can be evaluated by:

$$\begin{cases} \sigma_{top}(t) = \sigma_i(t) + \sigma_r(t) = \dfrac{A}{A_{si}} E_p[\varepsilon_i(t) + \varepsilon_r(t)] \\ \sigma_{bot}(t) = \sigma_t(t) = \dfrac{A}{A_s} E_p \varepsilon_t(t) \end{cases} \tag{5}$$

where A is cross-sectional area of incident and transmitted plate, A_s is cross-sectional area of specimen, A_{si} is cross-sectional area of specimen with cleavage triangle end, E_p is elastic modulus of the plates, ε_i (t), ε_r (t) and ε_t (t)are the strains recorded by strain gauges glued to the surface of incident and transmitted plate, respectively.

Fig. 3 shows the time-dependent loading curves for sandstone under different freeze-thaw cycles. It can be seen that the peak strength decreases with the increasing freeze-thaw cycles, and the peak strength is 40.08 MPa, 38.54 MPa, 32.79 MPa, 29.10 MPa, 28.31 MPa, and 27.84 MPa, respectively, indicating that rocks under cyclic freeze-thaw are more easily destroyed.

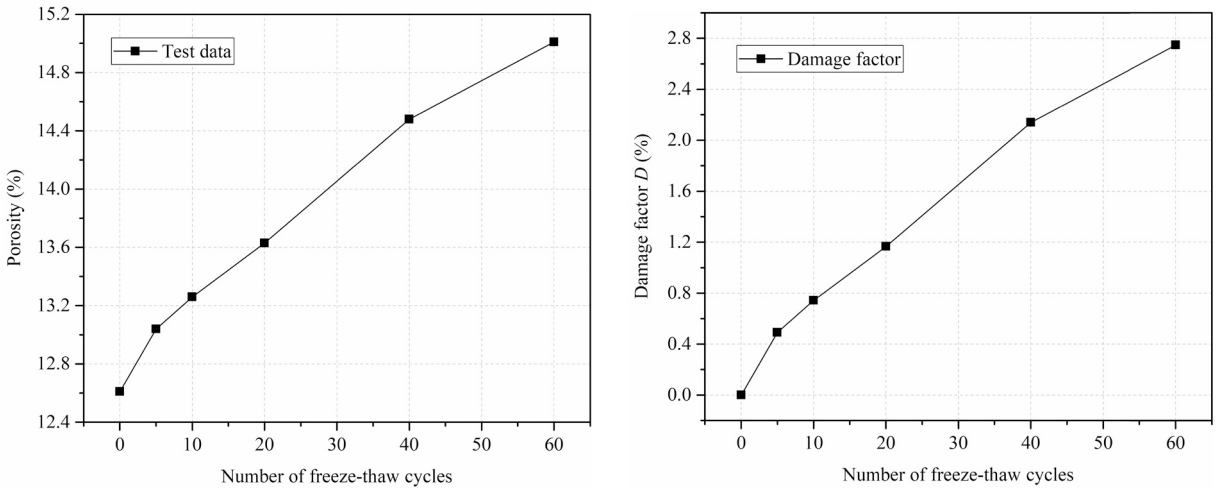
3.2. Changes of ultrasonic waves

The measurement results of P-wave and S-wave for treated specimens are presented in Table 1. According to Ravi-Chandar (2004), dynamic elastic modulus E_d, Poisson's ratio μ_d and Rayleigh wave velocity c_R can be determined by the following formulas:

$$E_d = \rho c_p^2 \frac{(1+\mu_d)(1-2\mu_d)}{1-\mu_d} \times 10^{-9} \tag{6}$$

Table 2
Spectrum area of specimens at various freeze-thaw cycles.

Spectrum area	Cycles					
	0	5	10	20	40	60
Area of the first peak	4990.99	5493.36	6566.91	6752.89	7818.26	8222.95
Area of the second peak	6714.74	7028.66	7744.18	8835.27	10,398.32	10,607.59
Area of the third peak	110.73	123.62	163.32	175.95	210.50	188.12
Total area	11,816.46	12,645.64	14,474.41	15,764.11	18,427.08	19,018.66



(a) Porosity (b) Damage factor

Fig. 6. Relations among porosity, damage factor and freeze-thaw cycles.

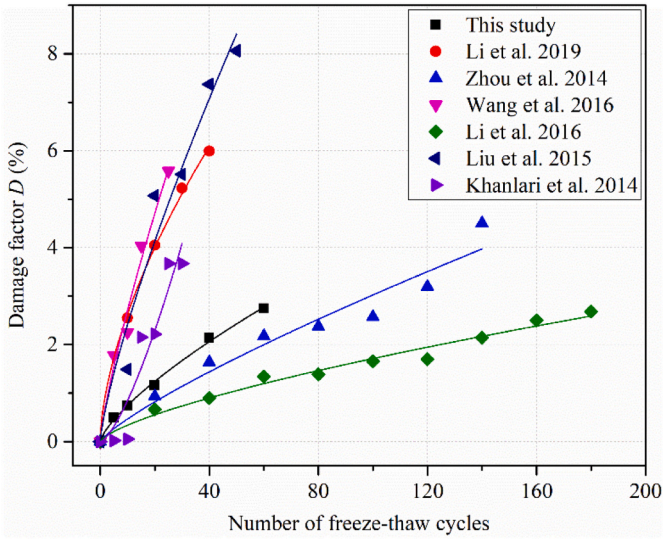


Fig. 7. Relation between damage factor and freeze-thaw cycles.

Table 3
Relations between damage factor and freeze-thaw cycles.

Researches	Rock	Initial porosity (%)	Relation	R^2
This study	Red sandstone	12.61	$D = 0.139t^{0.731}$	0.995
Li et al. (2019b)	Sandstone	9.30	$D = 0.655t^{0.604}$	0.993
Zhou et al. (2015)	Sandstone	7.57	$D = 0.073t^{0.810}$	0.894
Wang et al. (2016a)	Red sandstone	15.80	$D = 0.432t^{0.797}$	0.943
Li et al. (2016)	Yellow sandstone	6.53	$D = 0.068t^{0.702}$	0.961
Liu et al. (2015)	Red sandstone	22.86	$D = 0.406t^{0.775}$	0.900
Khanlari et al. (2014)	Sandstone	5.78	$D = 0.031t^{1.431}$	0.852

Table 4
Proportion of pore sizes (%).

Pore size	Cycles					
	0	5	10	20	40	60
Micropore	48.59	46.31	43.17	42.43	41.74	39.86
Mesopore	50.65	52.70	55.78	56.43	57.00	58.84
Macropore	0.76	0.99	1.05	1.14	1.26	1.30

$$\mu_d = \frac{(c_p/c_s)^2 - 2}{2\left[(c_p/c_s)^2 - 1\right]} \tag{7}$$

$$\left(2 - \frac{c_R^2}{c_s^2}\right) = 4\left(1 - \frac{c_R^2}{c_d^2}\right)^{1/2}\left(1 - \frac{c_R^2}{c_s^2}\right)^{1/2} \tag{8}$$

where ρ is density.

Substituting the measurement results of P-wave and S-wave velocity into Eqs. (1), (2) and (3), elastic constants E_d, μ_d and Rayleigh wave velocity c_R can be obtained, and average values were presented in Table 1.

The relation between acoustic velocities and freeze-thaw cycles are given in Fig. 4, one can find that both P-wave and S-wave velocities decrease with freeze-thaw. Compared with P-wave velocities, the reduction amplitude of S-wave velocities is smaller. After 60 cycles, the P-wave velocity and S-wave velocity decrease by 15.85%, and 12.34%, respectively. This is because the original pores expand due to the water-ice phase change and the uneven shrinkage and expansion of mineral grains, thus a large amount of water enters the pores of specimen, which further contribute to the expansion of pores. When the ultrasonic wave passes through the rock containing pores, the ultrasonic wave velocity will be attenuated due to the refraction, reflection and energy loss of ultrasonic wave.

3.3. NMR test results

3.3.1. T_2 distribution

Fig. 5 shows the T_2 spectrum curves of treated specimens. The size of pores can be determined by the x-axis T_2 value, and the number of pores depends on the spectrum area. One can find that T_2 spectrum distributions of sandstone specimens are trimodal, which reflects multiple pore types. Yao et al. (2010) pointed out that three spectral peaks correspond to the micropore, mesopore, and macropore, respectively. In Fig. 5, the T_2 value of the three peaks is about 0.90 ms, 16.25 ms, and 410.20 ms, respectively, and the amplitude of the first peak and the second peak are much larger than that of the third peak, indicating that pores inside the rock are mainly micropore and mesopore.

To quantitatively analyze the dynamic variation of pore volume, the spectrum peak area of the sandstone was calculated, as presented in Table 2. It can be observed that the T_2 spectrum peak area increases with freeze-thaw, which means that the micropores in the rock gradually expand to larger pores during the freeze-thaw process, and new micropores are generated during the process of water-ice phase transition.

3.3.2. Porosity and damage factor

The amplitude of the T_2 distribution curve is proportional to the amount of hydrogen protons contained in the pore fluids within the rock. Thus, this amplitude can be calibrated to give a porosity value (Coates et al., 1999). The procedures can be described as follows. By

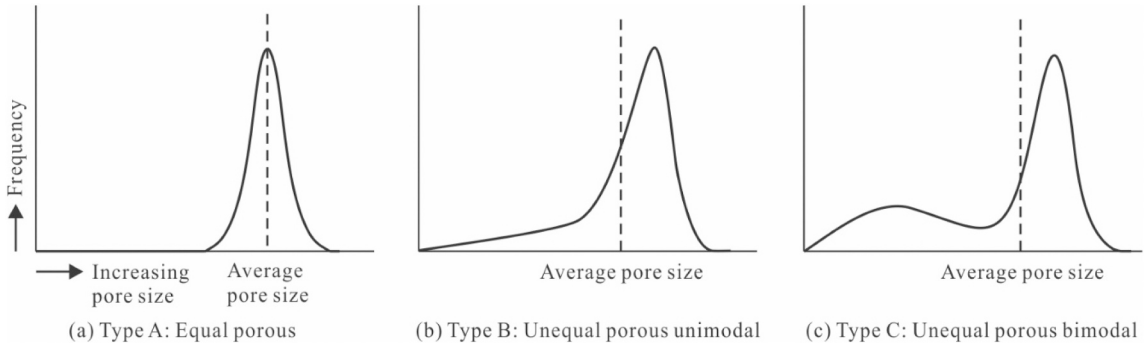
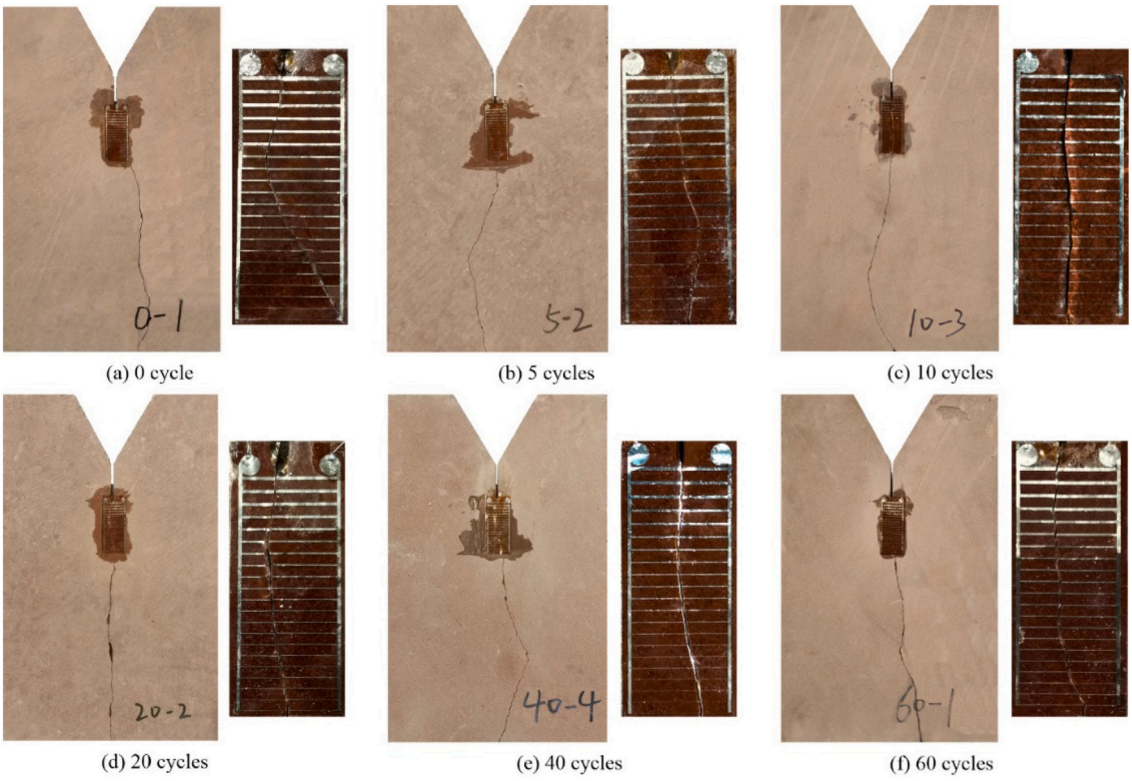


Fig. 8. Three pore types of sandstone according to the pore-radius distribution (A.C.G. et al., 2004).



(a) 0 cycle (b) 5 cycles (c) 10 cycles

(d) 20 cycles (e) 40 cycles (f) 60 cycles

Fig. 9. Experimental results of SCT specimens at various freeze-thaw cycles under impact loads.

testing a set of specimens with the known water content (porosity), a fitted curve, or a calibration curve, between water content and NMR amplitude can be determined. Putting measurement results into the calibration curve, the porosity of the specimen can be obtained.

Fig. 6(a) shows the relation between the porosity and cyclic freeze-thaw. Apparently, the porosity shows a non-linear relationship with freeze-thaw cycles, and the increase amplitude gradually decreases. After 60 cycles, the porosity is the greatest, which increases by 19.03% compared with specimens without freeze-thaw. This is because cyclic freeze-thaw can greatly contribute to the generation and expansion of pores. As the temperature rises, the ice in pores melts, and the water migrates continuously, which provides favorable conditions for the development of the frost heaving force during the next freezing period, and finally causes the increasing porosity. Similar trends can be observed by Liu et al. (2019) and Li et al. (2019b).

To further analyze the freeze-thaw damage degree of specimens, the damage factor (D) that relates to the porosity is established:

$$D = \frac{n_i - n_0}{1 - n_0} \times 100\% \tag{9}$$

where n_0 is the initial porosity of the specimen without freeze-thaw, and n_i is the porosity of the specimen after i freeze-thaw cycles.

Fig. 6(b) shows the relation between the damage factor and freeze-thaw cycles, which has a similar variation trend as Fig. 6(a). From 0 to 60 cycles, the damage factors of specimens are 0.49%, 0.74%, 1.17%, 2.14%, and 2.75%, respectively, indicating that the freeze-thaw damage is an accumulative process. Many scholars have conducted a large number of experiments to study the effect of freeze-thaw on porosity, but the fitting relation is not given (Khanlari et al., 2014; Li et al., 2016; Li et al., 2019b; Liu et al., 2015; Wang et al., 2019; Wang et al., 2016a; Zhou et al., 2015). For quantitative comparison, the relation between damage factor and freeze-thaw cycles are given in Fig. 7. Meanwhile, and Table 3 shows the fitting equations for rock under freeze-thaw cycles. It can be found that the relation between the damage factor and freeze-thaw cycles can be well fitted by the power equation except for the results obtained from Khanlari et al. (2014). As the number of freeze-thaw cycles increases, the damage always increases, and the damage degree varies greatly for sandstone under the same number of freeze-thaw cycles. The main reason comes from the difference in microstructure, mineral composition and initial porosity of sandstone. When the initial porosity of rock is relatively large, such as larger than 9.30%, the damage degree is serious. In addition, the freeze-thaw test method is also critical to the deterioration degree of the rock, e.g., the temperature range and the freezing rate.

3.3.3. *Pore structure*

The pore size distribution, or pore structure, has significant influences on mechanical characteristics of rocks (Zhou et al., 2020). According to Yao et al. (2010) mentioned in part 3.3.1, the proportion of three types of pore size for treated specimens are given in Table 4. One can find that the number of micropore and mesopore account for more than 98% of total pores. The proportion of mesopore and macropore increase with freeze-thaw, whereas the proportion of micropore decreases, indicating that more micropores are transformed into mesopore and macropore during the freeze-thaw process.

In a previous study (Ruedrich and Siegesmund, 2006), the sandstone pore size distributions are separated into three types, and the ability of sandstone to resist freezing and thawing is dominated by the pore size distribution. During the freeze-thaw experiment, sandstones with a unimodal pore size distribution (Type A) could complete more cycles than sandstones with a larger percentage of micropores (Type C), thus, the freezing and thawing resistance of Type A is the greatest, followed by Type B and Type C. (See Fig. 8)

However, the pore size distributions of rocks were obtained using mercury porosimetry method in previous study, which will affect the data quality. Due to the high-pressure intrusion of mercury, the skeletal porous structure of the specimen may be destroyed, and the measured number of large pores may be reduced because of the shielding effect of smaller pores (A.C.G. et al., 2004). In this study, it can be concluded from Fig. 5 that there are three peaks about pore size distribution of

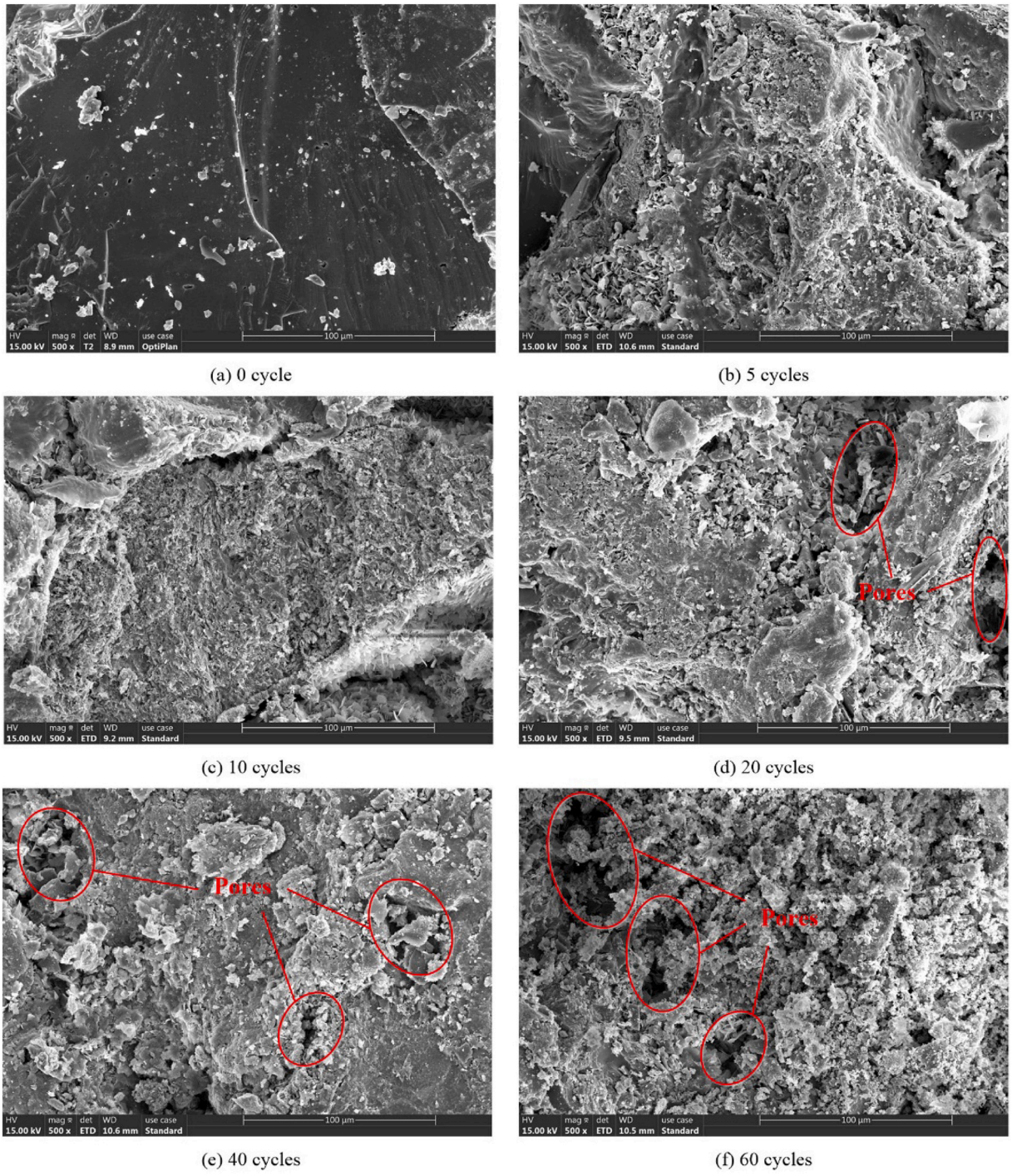


Fig. 10. SEM images of fracture surfaces of specimens under various freeze-thaw cycles.

sandstone. This is because NMR technique has the advantage of not destroying the specimen, thus all pores in the sandstone can be detected. Therefore, the sandstone used in this test can be classified as Type C, indicating that the freezing and thawing resistance of the red sandstone is weak and is more susceptible to weathering.

3.3.4. Physical mechanism of freezing-thawing process

The freeze-thaw cycle affects the temperature change and water migration of the internal structure of the rock. On the one hand, the uneven distribution of cracks and pores causes uneven temperature distribution. Affected by the temperature difference, the shrinkage of grains and the frost heaving force generated between the cracks lead to a stress concentration zone at the tip of the crack. When the strength caused by the stress concentration is higher than the yield strength of the rock material, the crack will propagate and failure. On the other hand, during the freezing process, the unfrozen water at the ends of the cracks increases the water pressure, which promotes the expansion of the pores and cracks. During the melting process, water replenishment and migration along the crack surface provide favorable conditions for the next icing and expansion. With repeated freezing and thawing, the cracks propagate deeply, and the local damage zone become significant. Finally, the macroscopic changes are mainly reflected in the increase in the number and size of pores and cracks, and the macroscopic changes are mainly manifested in the generation of surface cracks and the

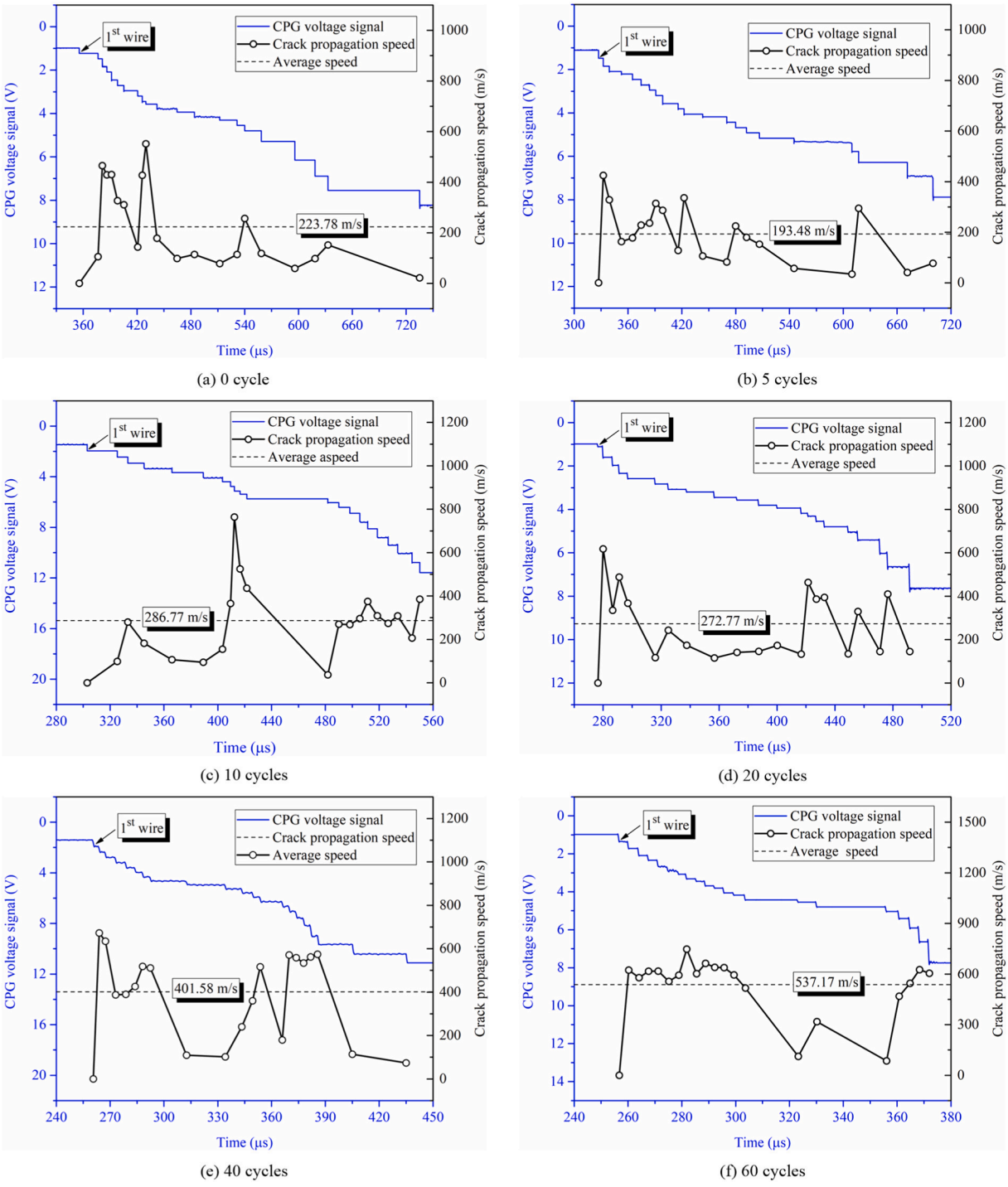


Fig. 11. Histories of voltage signals and crack velocities specimens under various freeze-thaw cycles.

shedding of cuttings.

3.4. Impact test results

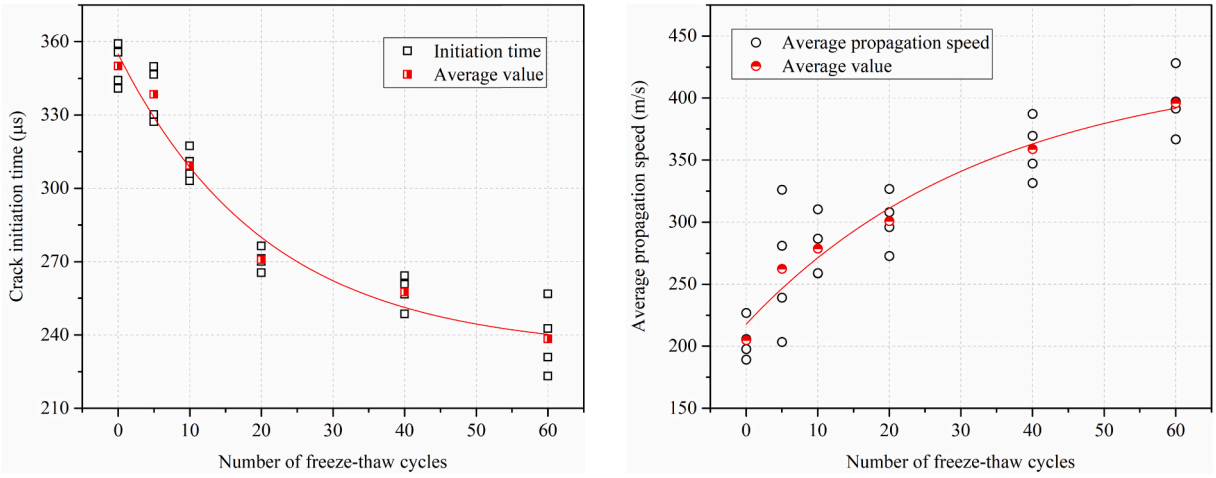
3.4.1. Fracture modes

The typical failure photos of SCT specimens are presented in Fig. 9. Generally, the crack propagation path for sandstone will be deflected as the crack length reaches a certain level, this is because the heterogeneity of sandstone material (see Fig. 1).

To further understand fracture mechanisms of treated sandstone specimens under dynamic loads, SEM measurements were performed to obtain the microstructures of fracture surfaces near crack tips, and Fig. 10 shows the microstructures magnified 500 times. From Fig. 10(a), the fracture surface of the sandstone without freeze-thaw is smooth, indicating that the grain destruction was mainly transgranular. This is because the cementing force between mineral grains are strong, the energy required for rock failure is large, thus, the grain destruction was mainly transgranular. Fig. 10 (b)-(f) are the surface morphology after freezing and thawing. It can be found that the fracture surface become rough and structure become loose. With the rise of freeze-thaw, the boundaries between crystal grains can be clearly observed, and the amount and size of pores gradually increase, indicating that the

Table 5
Fracture parameters of sandstone under different freeze-thaw cycles.

Specimen ID	Initiation time /μs	Ave. /μs	Max. velocity /(m/s)	Min. velocity /(m/s)	Ave. velocity /(m/s)	Ave. /(m/s)	DIT /MPa•m$^{1/2}$	Ave. /MPa•m$^{1/2}$
0–1	355.64	349.95	551.66	21.56	197.78	204.92	7.17	7.04
0–2	340.80		550.01	59.19	226.78		6.84	
0–3	359.12		345.24	42.02	189.26		7.21	
0–4	344.23		402.56	32.58	205.84		6.94	
5–1	327.24	338.43	424.71	34.09	193.48	262.48	6.20	6.54
5–2	330.12		397.11	41.64	281.08		6.27	
5–3	346.51		502.28	57.65	346.19		6.79	
5–4	349.85		394.27	34.02	229.16		6.89	
10–1	303.01	309.29	763.89	36.50	286.77	278.75	6.10	6.05
10–2	310.94		445.34	52.17	258.80		6.02	
10–3	305.95		415.09	79.54	259.00		6.07	
10–4	317.26		563.16	58.51	310.44		6.01	
20–1	276.43	270.81	617.98	115.30	272.77	300.94	6.06	5.85
20–2	271.26		458.33	26.78	296.09		5.88	
20–3	265.48		685.36	36.62	308.13		5.62	
20–4	270.06		723.88	41.48	326.77		5.84	
40–1	260.66	257.50	672.37	73.43	331.58	358.90	5.08	4.90
40–2	256.59		488.59	62.21	347.23		4.85	
40–3	264.22		706.94	57.06	369.56		5.27	
40–4	248.54		751.08	88.96	387.24		4.40	
60–1	256.75	238.35	747.29	85.11	397.17	395.88	4.56	3.64
60–2	242.53		589.81	55.03	366.76		3.82	
60–3	230.92		1139.90	97.64	428.12		3.28	
60–4	223.18		713.98	117.30	391.45		2.88	



(a) Crack initiation time (b) Average crack propagation speed

Fig. 12. Relations among crack initiation time, average crack velocity and freeze-thaw cycles.

cementation property between material grains was distorted. After 20 cycles, mineral grains getting more separate from each other and the porosity become large. In addition, the fracture surface is rough, indicating that the grain destruction of sandstone is mainly intergranular with the increase of freeze-thaw cycles, which dissipates less energy.

3.4.2. *Crack initiation time and crack velocity*

The time when the first wire of CPG breaks is crack initiation time. Under dynamic loads, the wire will be broken successively by the running crack, causing a series of step-changes in voltage signals, and crack velocity can be calculated using the wire breaking time and spacing between two adjacent wires. Fig. 11 shows the histories of CPG voltage signals and crack velocities for treated specimens, apparently, the crack velocities vary with time, which can be explained by the anisotropy of the sandstone material. After 0, 5, 10, 20, 40 and 60 cycles, the variation coefficient of the crack velocity is 0.703, 0.564, 0.570, 0.543, 0.478, and 0.313, respectively. Generally, the discreteness of the crack velocity of specimens without freeze-thaw is large as compared to the specimens subjected to freeze-thaw.

Table 5 listed the test results of fracture properties of treated specimens. Fig. 12 shows the relations among crack initiation time, average crack velocity and freeze-thaw cycles. In Fig. 12(a), the crack initiation time nonlinearly decreases with freeze-thaw. After 60 cycles, the crack initiation time decreases by 31.07%, indicating that the crack in the specimen is the easiest to initiate. In addition, the fitting curve seems to has a transition at the 20th cycle. The initiation time is more cycle-sensitive before the 20th cycle.

From Fig. 12(b), the crack velocity increases with freeze-thaw. After 60 cycles, the crack velocity is about 2 times that of the specimen without freeze-thaw, which demonstrates that the freeze-thaw exerts marked effect on the fracture process of sandstones due to the degradation of microstructures.

3.4.3. *Dynamic fracture toughness*

Dynamic fracture toughness include dynamic initiation toughness (DIT) and dynamic propagation toughness (DPT), which are determined using numerical-experimental method (Zhou et al., 2019; Zhou et al., 2018; Zhu et al., 2014). Fig. 13 shows the calculation model built by

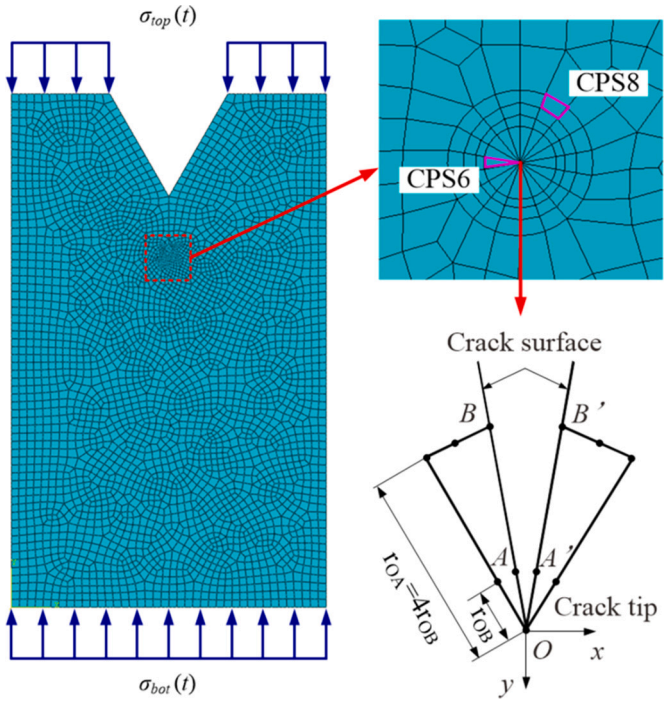


Fig. 13. The mesh and boundary conditions of the model.

ABAQUS code. In order to simulate singularity and improve calculation accuracy, the detailed focused meshes, 6-node triangular elements CPS6, were meshed for the crack front, and the other areas were meshed with 8-node quadrilateral elements CPS8. Totally 17, 642 elements and 53, 499 nodes were meshed for the SCT specimen. Applying the parameters shown in Table 1 (ρ, E_d, and μ_d) and loads given in Fig. 3 on the model, the stress intensity factor (SIF) can be obtained.

According to Rose (1976), the SIF under dynamic loads, or DSIF, can be computed as follows:

$$K_I^d(t) = k(v)\cdot K_I^0(t) \tag{10}$$

where $K_I^0(t)$ is the SIF of the stationary crack and $k(v)$ is universal function:

$$k(v) = \frac{1 - v/c_R}{\sqrt{1 - hv}} \tag{11}$$

where v is the crack velocity, and h is a function that relates to the elastic waves (c_p, c_s, and c_R):

$$h = \frac{2}{c_p}\left(\frac{c_s}{c_R}\right)^2\left(1 - \frac{c_s}{c_p}\right)^2 \tag{12}$$

Fig. 14 shows the DSIFs history of the specimen #60–1. For a crack at initiation, $k(v) = 1$ because the crack velocity is zero. From Fig. 14(a), the crack initiation time of the specimen #60–1 was 256.75 μs when the corresponding critical DSIF is 4.56 MPa•m$^{1/2}$, i.e., DIT of red sandstone. The results about the DIT of specimens are presented in Table 5.

For a moving crack, such as the crack at the 6th wire, the DSIF $K_I^d(t)$ history was obtained by the Eqs. (10) and (11), as shown by the red curve in Fig. 14(b). The 6th wire breaking time was 275.17 μs, the crack velocity was 556.68 m/s, $k(v)$ was equal to 0.59, and the corresponding critical DSIF is 3.70 MPa•m$^{1/2}$, i.e., DPT of red sandstone.

Similarly, we can get the critical DSIFs for specimen #60–1 at other wire. The comparison of critical DSIFs of six typical treated specimens are depicted in Fig. 15. It can be concluded that cyclic freeze-thaw has a considerable deteriorative effect on the sandstone fracture toughness, specimen without freeze-thaw generally has the highest DPT, and the

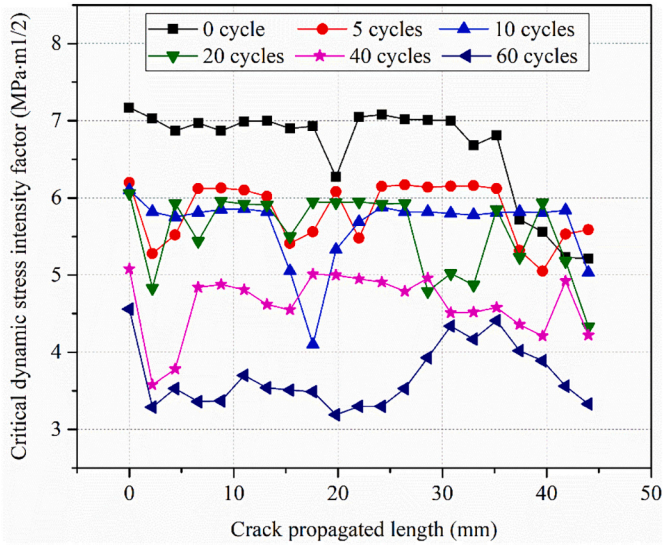
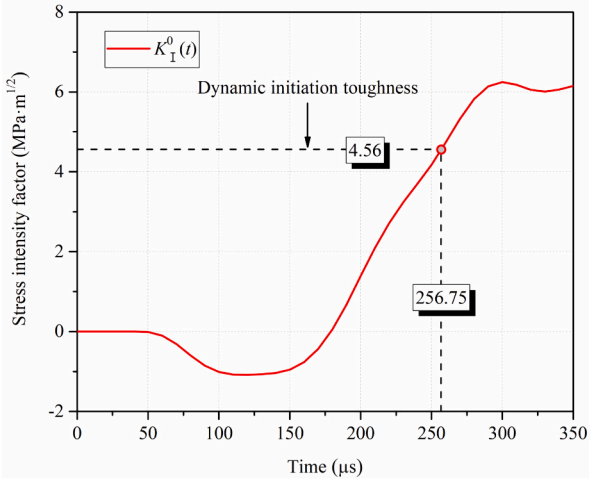
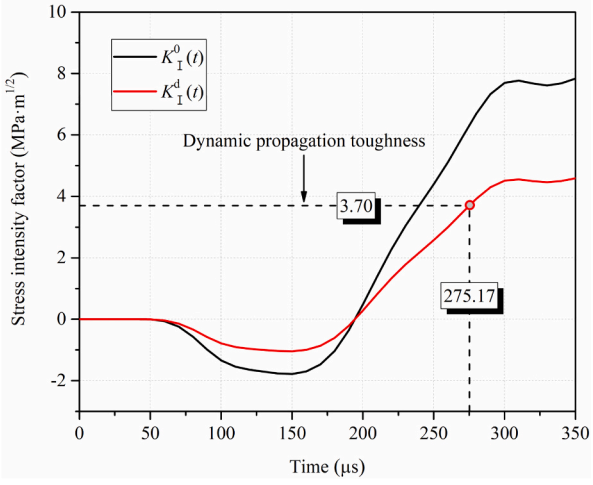


Fig. 15. Relation between critical DSIFs and crack length.

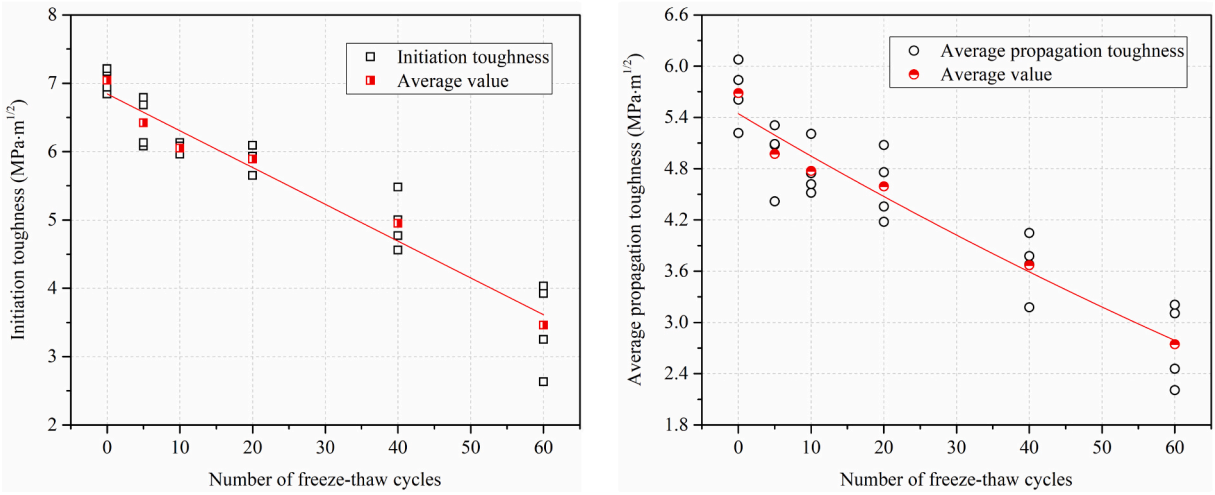


(a) Dynamic initiation toughness



(b) Dynamic propagation toughness of crack at the 6th wire

Fig. 14. The critical DSIFs histories for the specimen under 60 freeze-thaw cycles.



(a) Initiation toughness (b) Average propagation toughness

Fig. 16. Relations among DIT, average DPT and freeze-thaw cycles.

specimen after 60 cycles has the lowest DPT. The DSIF varies with the crack propagated length, and there is a jump in some stages, which is related to the heterogeneity of the material. Additionally, the critical DSIF at propagation is generally less than that at initiation. This is because the kinetic energy was generated by the moving crack.

Fig. 16 shows the relations among the DIT, average DPT and freeze-thaw cycles. It appears that both them decrease with freeze-thaw. After 60 cycles, the DIT and the average DPT decrease by 48.37%, and 48.45%, respectively. The reduction of fracture toughness is related to freeze-thaw action which can greatly contribute to the attenuation of solid skeleton, this can be confirmed from Fig. 10.

4. Conclusions

To investigate the microstructure changes and dynamic mechanical characteristics of sandstone subjected to freeze-thaw, laboratory freeze-thaw tests were conducted on SCT specimens first, and then DPI tests were performed using treated SCT specimens. P-wave, S-wave, crack propagation parameters and fracture toughness were determined, the microscopic damage characteristics of sandstone specimens were investigated using NMR technique, and the microstructure of the fracture surface was analyzed by SEM. From this study, several conclusions can be obtained.

(1) Cyclic freeze-thaw has a deteriorative influence on the rock microstructure. Red sandstone has three different pore types, i.e., micropore, mesopore, and macropore, and micropore and mesopore play a dominant role in damage deterioration during freeze-thaw process.
(2) CPG test results indicate that the crack initiation time shows a non-linear reduction with freeze-thaw cycles, and the increase amplitude gradually decreases. The crack propagation velocity increases with freeze-thaw cycles.
(3) The grain destruction of the sandstone without freeze-thaw is mainly transgranular, whereas the fracture mode is mainly intergranular after 60 cycles.
(4) The dynamic fracture toughness decreases with freeze-thaw cycles, after 60 cycles, its only about half of that of specimens without freeze-thaw.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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